

Sensory Feedback in a Serious Gaming Environment and Virtual Reality for Training Upper Limb Amputees

Reidner Santos Cavalcante ^{1*}, Edgard Lamounier ¹, Alcimar Soares ¹, Aya Gaballa ², John-John Cabibihan ²

Federal University of Uberlândia, Uberlândia, Brazil¹; Qatar University, Doha, Qatar²

ABSTRACT

In this article, the authors present a system based on Immersive Virtual Reality and Serious Games for training the use of prostheses by upper limb amputees with a tactile feedback. By using EMG signal processing, users can control the opening and closing of a virtual prosthesis, just like in real life. Tactile feedback causes an improvement in the sensation of touch. Tests were carried out with separate groups: with and without sensory feedback with amputated and non-amputee volunteers. Tests in which users who received haptic feedback demonstrated improvements in performance compared to those who did not use haptic feedback.

Keywords: Immersive Virtual Reality, sensorial feedback, workload, serious game.

Index Terms: K.4.2. [Computers and Society]: Social Issues- Assistive technologies for persons with disabilities.

1 INTRODUCTION

The amputation of a limb is a traumatic event capable of triggering major changes in all areas of the subject's life [1]. The final fitting of the prosthesis, along with the required therapeutic and training steps, can take many months to complete, requiring significant effort on the part of the amputee.

In many cases, adapting to the prosthesis will be mentally and physically tiring, requiring considerable time over several weekly sessions; and the time required and low motivation observed in traditional methods are often reported as reasons for abandoning the rehabilitation program and the main causes of therapeutic failure [2].

Serious Games (SG) are becoming popular in various learning methods such as healthcare, education, and training [3]. Most Virtual Reality (VR) systems reported in the upper limb prosthetics training literature are based on non-immersive environments and lack of sensory feedback [4]. Unfortunately, such an approach cannot deliver the natural and intuitive interactions claimed by most works.

Furthermore, providing only visual feedback drastically limits the immersive experience for prosthetic training purposes, as it requires a great deal of spatial vision for users to reach and manipulate various objects during computer-aided training.

Current devices will require users to generate non-intuitive and continuous co-contractions of the remaining stump muscles [5]. In this particularly challenging task, users can use vibration, torque, and even auditory feedback in the 'real world' to sense when the prosthesis grips an object and when slipping occurs [6]. Based on this information, this article presents the development of an immersive virtual reality (iVR) training environment for the

rehabilitation of amputated upper limbs based on Serious Games with tactile feedback.

2 MATERIALS AND METHODS

The main components of the system are: HTC Vive Pro, Vive Tracker, EMG (electromyographic) controller (data acquisition and pattern recognition), vibrational armband, Myo Armband and a computer. The user interacts with the virtual environment by contracting the muscles in the stump and moving the arm. In the case of non-amputee volunteers, the forearm muscles are used.

The Vive Tracker, located on the arm, provides information so that the virtual prosthesis moves within the virtual environment as the user moves the arm. The EMG module processes contractions of the stump (or forearm) muscles and sends setpoint values to the virtual environment to indicate how much the virtual prosthesis should open/close the hand. The vibrational armband and EMG sensors are also positioned on the user's forearm or stump. Whenever the user grasps an object in the virtual environment, vibrational feedback is used to convey the sensation of touch.

This study was conducted in two distinct phases. In the first phase, tests were carried out with non-amputee volunteers with the aim of evaluating the system's functioning. This phase is crucial, as it can be frustrating for an amputee to use a system with many operating errors. In the second phase, tests were conducted with an amputee user to evaluate both the performance of the amputee user and the usability of the system.

Thus, the first phase included checking the system's functionality, identifying possible technical problems, and analyzing the performance of non-amputee users. For this phase, eight non-amputee male volunteers, aged between 20 and 40 years, were recruited. The volunteers were divided into two groups: G1 - performed the Box & Blocks (B&B) tasks while receiving visual and vibrational feedback; G2 - performed the B&B tasks receiving only visual feedback, without tactile feedback.

In this experiment, we conducted four trials using the Box and Blocks environment, with the experimental group receiving haptic feedback. No tactile feedback was given to the control group. In each trial, volunteers were asked to move all the blocks from one side of the box to the other as quickly as possible. The time for each trial was recorded.

In the second phase of the study, carried out after the improvements identified in the first phase, tests were specifically conducted with an amputee user, Figure 1. This approach allowed us to evaluate the usability of the system in a more realistic context, considering the specific needs and challenges faced by these users. The tests included evaluating the system's usability, effectiveness in carrying out tasks, and overall user satisfaction.

At this stage, a volunteer with a 30-year-old transradial amputation was selected. The volunteer performed eight tests in the B&B environment without vibrational feedback and eight tests with vibrational feedback. The Myo bracelet was used to capture myoelectric signals and provide vibrational feedback. The SUS questionnaire was applied to evaluate the usability of the system, and the NASA-TLX was used to evaluate the workload, in addition to the data collected by the system.

* reidner.cavalcante@gmail.com



Figure 1: Amputee volunteer with Myo Armband and HTC Vive Pro positioned.

3 RESULTS AND DISCUSSION

Figure 2 shows the average completion time of all attempts performed by non-amputee volunteers in the first phase of the study in groups G1 and G2, respectively. At first, both groups faced greater difficulty, taking longer time to complete the tasks while adapting to controlling the virtual prosthesis using myoelectric signals.

As the trials progressed, especially for the G1 volunteers, completion time decreased steadily. Although G1 started with a considerably higher average value than G2, due mainly to the time spent by G1-V3 (Volunteer 3) on the first attempt, the group that received tactile vibrational feedback showed better overall performance. On average, G2 took 2.3 times longer to complete the task on the fourth attempt.

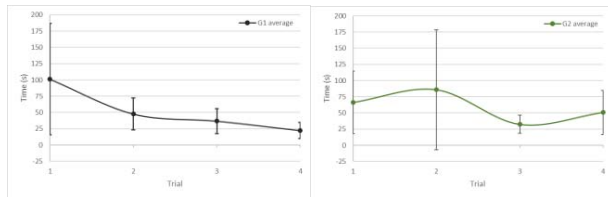


Figure 2: Average completion time per Test for group G1 and G2, respectively.

In the second phase of the study, conducted with an amputee volunteer, we gathered data on the successful movement of objects to the opposite side of the box. Figure 3 illustrates these findings, categorizing the tests based on whether tactile feedback was present or absent. Initially, the user achieved relatively low scores, indicative of their initial challenges. Notably, there was a substantial improvement in the accuracy of scores across all trials, regardless of the presence of tactile feedback.

However, in tests where tactile feedback was integrated, the user consistently achieved scores greater than or equal to 75% of the cases. This suggests that tactile feedback played a significant role in assisting the user, leading to enhanced performance in the tasks and more effective control of the virtual prosthesis.

4 CONCLUSION

In this study, a virtual training environment for upper limb amputees based on serious games and immersive Virtual Reality was presented. The system allows the user to train how to use a myoelectric prosthesis. Furthermore, a tactile vibrational feedback system was provided to better conduct user training.

Tests were carried out in the Box and Blocks environment. Although both groups in the first phase showed improvement in control of the virtual myoelectric prosthesis throughout testing,

participants who used haptic vibrational feedback showed better performance as testing progressed, compared to the group that did not use haptic feedback. In the second phase of the tests, it was noted that in tests using tactile feedback, the volunteer obtained higher scores than in tests without tactile feedback.

From the data collected using the NASA-TLX questionnaire, it was possible to identify that physical demand (40 points) and effort (60 points) are the factors that most affect the workload when performing tasks in this virtual environment, while temporal (10 points) and mental demand (10 points) have a low influence.

These results can be justified due to the control that the user must exert to keep the virtual prosthesis closed while moving the block. The volunteer exerted greater effort on the muscle used for capturing the EMG signal, aiming to prevent the prosthesis from performing the action of opening.

ACKNOWLEDGMENTS

The authors would like to thank the support and funding provided by the *Coordenação de Aperfeiçoamento de Pessoal de Nível Superior*– Brazil (CAPES) and the University of Qatar under grant IRCC-2019-001.

REFERENCES

- [1] Galván, G. B., & Amiralian, M. L. T. M. (2009). Corpo e identidade: reflexões acerca da vivência de amputação. *Estudos de Psicologia (Campinas)*, 26, 391-398..
- [2] Sharma, A., Hunt, C. L., Maheshwari, A., Osborn, L., Lévy, G., Kaliki, R. R., ... & Thakor, N. (2018). A mixed-reality training environment for upper limb prosthesis control. In 2018 IEEE Biomedical Circuits and Systems Conference (BioCAS) (pp. 1-4).
- [3] Muñoz, M., Pacheco, M. Á. G., & Reveles, J. G. H. (2021). Biblioteca de juegos serios para hacer más efectiva la enseñanza de Kanban acorde a objetivos de aprendizaje enfocados. *RISTI - Revista Ibérica de Sistemas e Tecnologías de Informação*, (41),1-16.
- [4] Melero, M., Hou, A., Cheng, E., Tayade, A., Lee, S. C., Unberath, M., & Navab, N. (2019). Upbeat: augmented reality-guided dancing for prosthetic rehabilitation of upper limb amputees. *Journal of healthcare engineering*, 2019.
- [5] Mattioli, F. E., Lamounier, E. A., Cardoso, A., Soares, A. B., & Andrade, A. O. (2011). Classification of EMG signals using artificial neural networks for virtual hand prosthesis control. In 2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society (pp. 7254-7257). IEEE.
- [6] Cavalcante, R. S., Júnior, E. A. L., Cardoso, A., Soares, A., & de Lima, G. M. (2018). Development of a serious game for rehabilitation of upper limb amputees. In 2018 20th Symposium on Virtual and Augmented Reality (SVR) (pp. 99-105). IEEE.